

HealthConnect Clinic Experience Lab: Integration Report

Week 8 Multidisciplinary Deliverable — Data Analytics to Data Science Track

1. EXECUTIVE SUMMARY

This report documents the formal output exchange between the Data Analytics and Data Science tracks. By transitioning from isolated workstreams to active multidisciplinary exchange, analytical insights regarding appointment friction (\$N = 5,000\$, 48.46% no-show rate, 29.64-day mean lead time) directly informed predictive feature selection and model interpretation.

2. MANDATORY 9-POINT INTEGRATION DOCUMENTATION

Track Collaborated With	Data Science Track
Dependency	Data Science required exploratory insights and baseline feature identification to properly scope predictive modeling targets and reduce feature dimensionality.
Output Received	Model feature requirements, target variable specifications, and baseline error distribution tolerances.
Output Provided	Validated no-show patterns, establishing a 48.46% no-show rate and a 29.64-day mean booking lead time baseline.
Final Integration Activity	Cross-checked statistical descriptive findings and distribution curves directly against the machine learning model's output feature importance scores.
What Changed	Shifted predictive feature selection away from generic demographic variables toward high-impact temporal decay thresholds (specifically isolating booking lead times exceeding 14 days).
Improvement Realized	Removed uninformative noise from the training dataset, increasing model predictive accuracy and ensuring close alignment with real-world clinical behavior.
Evidence	Analytics-to-Science Feature Alignment Tables & Exploratory Correlation Heatmaps.
Presentation Contribution	Provides the foundational narrative explaining how empirical data findings directly drove predictive model design and executive interpretation.

3. VERIFICATION ARTIFACTS & EVIDENCE

Empirical Cross-Track Verification

- Analytics-to-Science Feature Alignment Tables:** A cross-reference matrix matching statistical correlation coefficients against algorithm feature weights (validating lead time and historical attrition as primary drivers).
- Exploratory Correlation Heatmaps:** Visual covariance matrices mapping booking lead times and historical attendance directly to patient default probability.